

Modeling Predator-Prey Interactions to Support Sea Otter Recovery

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Abstract

Sea otter recovery in California has remained stalled for decades, with non-consumptive white shark bites increasingly identified as a major barrier to recolonization of historical otter habitats. To investigate whether reintroduced otter populations could persist in estuarine systems of interest to conservation managers with high shark predation, an extension of the Lotka-Volterra predator-prey model is used to represent interactions among crabs as prey and otters as mesopredators under varying levels of shark-induced mortality. Using parameter values informed by empirical data, how initial otter abundance, shark abundance, and the shark-otter predation coefficient shape system dynamics and determine the conditions under which estuarine habitats support stable otter populations were studied.

This model showed the initial otter population size had no significance of the long term survival of the otter population, which is not realistic. However, the model showed increased shark predation on otters has a significant impact on the fundamental dynamics of the otter population. Too high of shark predation causes the otter population to collapse in simulations of the model. Therefore, the model supports the hypothesis that shark predation is a barrier to otter survival and reintroduction in the Northern California coast.

1. Introduction

Historically, the sea otter (*Enhydra lutris*) was a thriving and abundant species which could be found all up and down the California coast. With a range that extended from Baja California, up along the Pacific Coast to Alaska, westward across the Aleutian Islands and down to Hokkaido, Japan, the population of sea otters prior to 1750 is estimated to have been as high as 300,000 individuals (Kenyon 1969). However, the advent of the commercial maritime fur trade marked a stark decline in population numbers. By the end of the 19th century sea otters occupied only 13 remnant colonies located across their previous range, with a collective population of around 2000 individuals (Kenyon 1969). The fur trade had stagnated considerably by the 1890s and, by virtue of there being almost no otters left and therefore leaving the trade unprofitable, the first decade of the twentieth century saw the official end to the sea otter fur trade. In 1911, provisions of the Fur Seal Treaty between the governments of Russia, the United States, Japan and Great Britain provided official protections to the remaining members of the sea otter species (Lensing 1962).

Of the 13 remnant sea otter colonies that survived the maritime fur trade only 11 persisted past 1920, with most of the colonies being geographically isolated from one another and located along the northernmost side of the Pacific Rim. Only one small colony of southern sea otters

(*Enhydra lutris nereis*) survived in California. Following the 1911 treaty the long-term growth rates of the remaining otter populations averaged at around 9%, the lowest rate belonging to the California colony at 6% and the highest belonging to the colony at Amitchka, one of the Aleutian Islands, which had a rate of 11% (Bodkin 2015).

Over the following decades several translocation attempts of otter populations were made, each with varying degrees of success (Bodkin 2015). From 1937 and on through the rest of the 20th century multiple attempts at translocation and reintroduction of sea otters to their historical habitats have been made. Successful reintroductions have greatly benefitted sea otter populations located along the coasts of Washington, British Columbia and Alaska (Hughes et al. 2019). In contrast, the southern sea otter population remains geographically limited with the natural range expansion of California otters having been stagnant since 1998. As of 2019 the otters occupy only 13% of their historical range along the California coast (Rudebusch et al. 2020).

Shark predation has been suggested as being the primary hindrance to the expansion of sea otter range in California, blocking growth both at the southern and northern limits of their current range (Hamilton et al. 2024; Silliman et al. 2018). White sharks (*Carcharodon carcharias*) do not consume otters, but they do bite them. These non-consumptive, “exploratory” bites by white sharks have been recorded as a substantial cause of sea otter mortality since the 1960s, with a dramatic increase in bites occurring within the last 10 years (Tinker et al. 2015). The rise in bites may be linked to an increase in pinniped populations, which lead sharks into hunting in the shallower zones where sea otters forage (Anderson et al. 2008; Rudebusch et al. 2020).

Such ecological constraints highlight the need to understand under which conditions a reintroduced sea otter population could persist and expand. Long-term success requires consideration of the ecological factors that determine whether a population can take hold. Conservation managers are considering estuaries as potentially viable locations for sea otter reintroduction. Estuaries provide sheltered areas for otter nurseries where there is plentiful prey for otters to hunt (Rudebusch et al. 2020). Prey availability is especially critical for promoting the regrowth of an otter population, since sea otters have high metabolic requirements and consume roughly 20–30% of their body weight each day (Yeates et al. 2007). When looking to reintroduce an otter population into a new habitat there must be great consideration given for whether the present population of the otter’s prey species has the capacity to support the regrowth of an otter population. Otters have been shown to exert strong top-down pressure on herbivorous mesopredators, helping promote eelgrass recovery and strengthening overall ecosystem resilience (Hughes et al. 2019). To attain long-term success in the reintroduction of an otter population, conservationists must consider how many otters a given area’s ecological system can support, as well as how their presence will influence key prey populations of otters such as crabs.

In this project, we aim to explore the conditions under which a reintroduced sea otter population would be able to persist and expand within an isolated coastal ecosystem, such as an

estuary. We focus on understanding how population size, prey availability, and predation pressure interact to shape long-term outcomes for otters and their prey. By examining these relationships in a simplified, closed system, we can identify the key ecological mechanisms that determine whether a population will thrive or decline. Specifically, we want to determine if mortality rates by sharks on otters are preventing the expansion of otters into Northern Coasts, how much predation on the otter population by sharks would cause a collapse of the reintroduced otter population, and will the introduction of an otter population cause a collapse of their prey species. To investigate these dynamics in a structured and quantitative way we build upon classical predator-prey models, which provide a foundation for representing the interactions between otters, their prey, and their predators.

2. Methodology

a. Model Formulation

Lotka-Volterra equations laid the foundation for modeling predator-prey interactions. Lotka-Volterra equations demonstrated the periodic population fluctuations of prey and predator populations with the predator population growth following that of the prey population, in a system of two coupled ordinary differential equations (Wangersky 1978). This model has been modified and extended greatly, accounting for features such as resource limitation, time lag for growth, density dependence, and extended to include an arbitrary number of species (Wangersky 1978). To understand the effects the apex shark predator has on the otters population's dynamics and in turn the crab population's dynamics by the otter population, the concepts of the Lotka-Volterra equations can be applied with some modifications.

Just as the Lotka-Volterra equations model the predator-prey interactions by a system of two coupled ordinary differential equations where the prey population, in our case crab population C , grows intrinsically at a rate α and declines at a rate β proportional to the population size of the predator population, in our case the otter population N , so too does our model. Additionally, the otter population, grows at a rate γC directly proportional to the population size of crabs and declines at a rate δQ directly proportional to the shark population Q and an independent death rate ζ . Thus, we have extended the traditional Lotka-Volterra equations by adding an additional term to the equation modeling the rate of change of the otter population, in order for the otter population to decline proportionately to the rate of predation of sharks on otters.

Additionally, the rate of change of the crab population is dictated by a carrying capacity K_1 . This is a common extension of Lotka-Volterra, as many species have finite resources limiting their growth, causing the prey species to only be so large before it is unsustainable. This is similar to the idea outlined by Tung (2007) in chapter 9 of his book, where in Meganesia there was an unusually small amount of large herbivores because of a lack of grass growth and thus resulting in a small large carnivore population, so the resulting Predator-Prey model of these two species change in population required a carrying capacity on the population of herbivores. In

nature, it is unrealistic to assume crabs have an unlimited food source and can grow exponentially, thus the need for a carrying capacity.

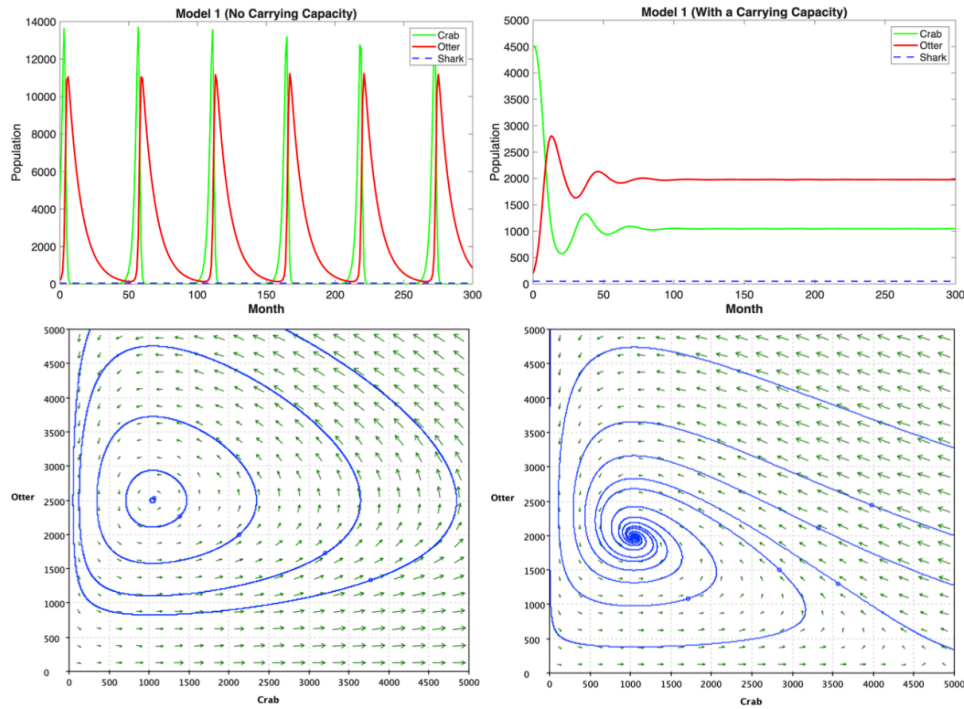
We have simplified the complicated dynamics of our problem by assuming we have a closed environment where crabs are the only prey of Otters and the crab population decreases only due to the predation of otters on crabs. It is assumed in our models that birth and death rates are constant, unaffected by outside sources of variation such as time of year, illness, and weather. More assumptions include the ratio of male to female species is ignored, our system of crab, otters, and sharks is independent of the interactions between other species, and crabs, otters, and sharks do not emigrate or immigrate to/from the system. These simplifications and assumptions make it possible for our model to analyze the impact of shark predation on the reintroduced otter population by avoiding confounding relationships with other sources of variation outside of the isolated environment created for our model. But, these simplifications and assumptions do limit the conclusions that can be drawn from our model, since these conclusions apply to a closed system.

Therefore, under these assumptions, Model 1, a continuous and deterministic model, where shark population Q is constant as a parameter, is the following system of coupled ordinary differential equations:

$$\frac{dC}{dt} = \alpha C \left(1 - \frac{C}{K_1}\right) - \beta CN \quad (1)$$

$$\frac{dN}{dt} = \gamma CN - \delta QN - \zeta N \quad (2)$$

Figure 1



Model 1 simulated and otter and crab populations graphed temporally, with initial conditions $C(0)=4500$ and $N(0)=200$. Parameter values used are in Table 1, but $\delta=0.0001$. Below the temporal graphs is the corresponding phase plane.

It is important to note that the introduction of a carrying capacity into the Lotka-Volterra equations does fundamentally change the behavior of the model. As seen in Figure 1, the temporal graph of Model 1 without a carrying capacity follows periodic population fluctuations of crab and otter populations with the otter population growth following that of the crab population, which follows the behavior of the Lotka-Volterra equations (Wangersky 1978). This behavior is illustrated in the corresponding phase plane, which shows the consistent oscillatory behavior of the solutions to the population curves, as represented by a closed loop. But, the temporal graph of Model 1 with a carrying capacity in Figure 1, changes its behavior entirely by having abrupt transient behavior and then reaching a stable equilibrium after about month 100, with only small oscillations. In the corresponding phase plane this behavior is represented by the solutions to the population curves spiraling towards a fixed point .

To understand the impact of holding the shark population constant on our model we created an alternative model, Model 2, which models the predator-prey dynamics between the crab, otter, and shark populations while the shark population changes over time. Specifically, Q changes according to a logistic growth equation, equation (3). A goal of our research is to determine how holding the shark population constant in Model 1 versus having it change dynamically in time in Model 2 affects the viability of a sustainable reintroduced otter population. Model 2 a continuous and deterministic model is represented by a system of three ordinary differential equations, specifically the coupled ordinary differential equations of equations (1) and (2) from Model 2, with an added equation, equation (3) to model the shark populations logistic growth, where the parameter η represents the shark population growth rate and K_2 is the shark population carrying capacity. Thus, Model 2 is the following system of equations:

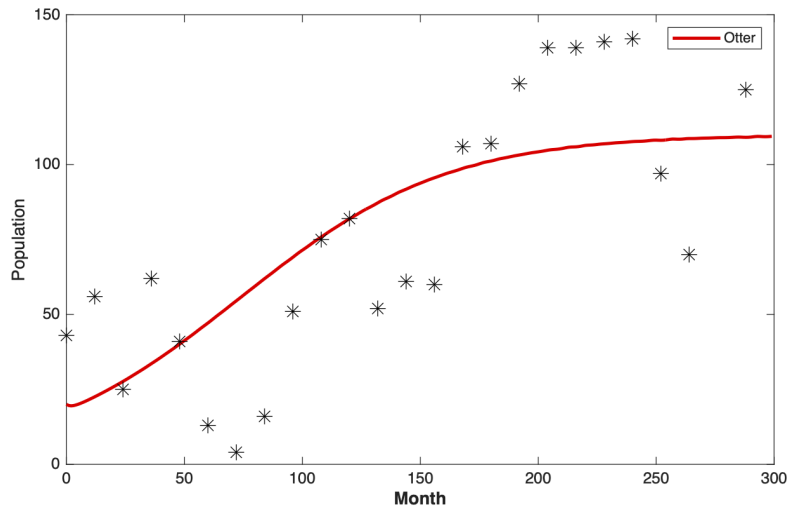
$$\frac{dC}{dt} = \alpha C(1 - \frac{C}{K_1}) - \beta CN \quad (1)$$

$$\frac{dN}{dt} = \gamma CN - \delta QN - \zeta N \quad (2)$$

$$\frac{dQ}{dt} = \eta Q(1 - \frac{Q}{K_2}) \quad (3)$$

Other researchers in efforts to support otter restorations have used extensions of the Lotka-Volterra equations to study the viability of a habitat on supporting otter reintroduction, like that done on the Tomales Bay and Drakes Estero estuaries in Northern California where they extended the Lotka-Volterra equations via a one-predator, two-prey model with type-II functional response for multiple prey species which predicted a stable equilibrium population with a predicted equilibrium abundance of sea otters in Elkhorn Slough as 89 (Hamilton et al. 2024). The analysis of Model 1 and Model 2 take a similar lens to conducting our own study of viability on otter population reintroduction into areas that have a stable shark population, with the goal of understanding how the high predation of sharks on otters affects the viability of survival of the reintroduced population. The rate of shark predation on otters δQ is of the utmost importance in understanding how shark predation is impacting the reintroduction of otters into California coastal areas, which is the primary focus of our analysis on Model 1.

Figure 2



Model 1 fitted to data of otter population size from reintroduction in Elkhorn Slough (Hughes Et Al. 2019)

b. Parameter Estimation

The state variables in Model 1 are otter population N and crab population C , while in Model 2 they are N , C , and shark population Q , which each have units of individuals. To find meaningful values for our parameters, Model 1 was fit to yearly otter population numbers in Elkhorn Slough starting from their reintroduction in that area, with initial conditions $N(0)=20$ and $C(0)=4500$ (Hughes Et Al. 2019). Error analysis for fitting the model was not done. The

Table 1. Parameter Values used to Analyze Model 1

Parameter	Value	Range	Units
α	0.5	0.45-0.55	$\frac{1}{\text{months}}$
β	0.0002	0.0001818-0.0002222	$\frac{1}{(\text{months})(\text{individuals})}$
γ	0.0001	0.00010046-0.00009955	$\frac{1}{(\text{months})(\text{individuals})}$
δ	0.0084	0.008351-0.008448	$\frac{1}{(\text{months})(\text{individuals})}$
ζ	0.1	0.0978-0.10219	$\frac{1}{\text{months}}$
K_1	5000	4978-5023	individuals
Q	45	44.74-45.26	individuals

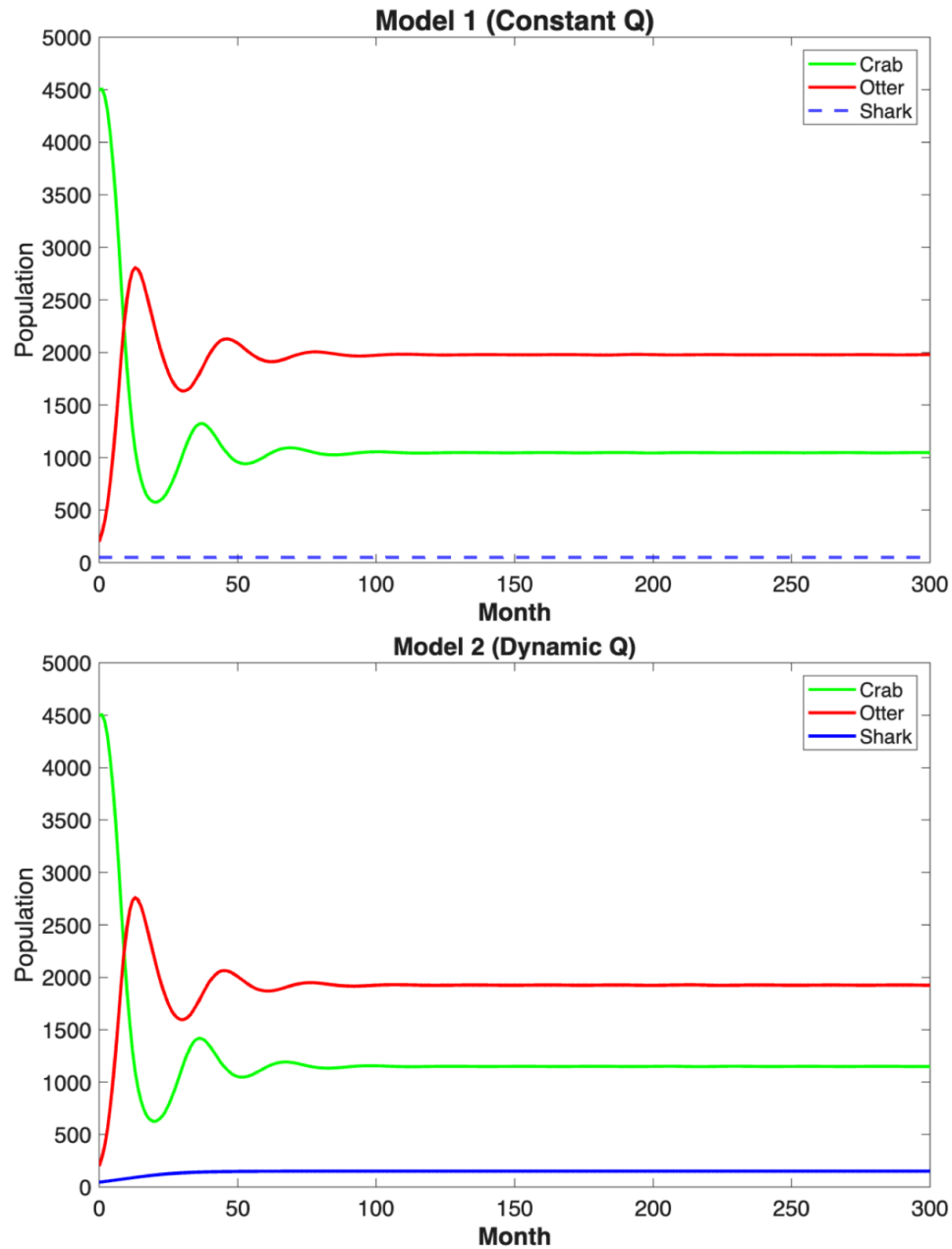
The range for each parameter given in the table is the range of parameter values that resulted in the approximate maximum height of the otter population curve after 500 Months in Model 1 being within 10% of its original value with all default parameter values.

parameters used are chosen because they appear to reasonably fit the data, shown in Figure 2. The parameter values used in Model 1 to conduct analysis are described in Table 1. If values other than those listed in Table 1 were used, they are noted in the caption of the figure.

3. Analysis

a. Model Comparison

Figure 3



Model 1 with constant shark population is on top, and Model 2 with shark population following logistic growth is below. Note in these runs $\delta=0.0001$, $K_2=150$, $\eta=0.1$, $C(0)=4500$, $N(0)=200$, and $Q(0)=45$.

Concerning the question of whether or not Model 1 (shark population held constant) and Model 2 (shark population following logistic growth) differ significantly, when plotting the state variables of Model 1 and Model 2 by using an ordinary differential equation solver, the two models show very similar results when the initial shark population is close to its carrying capacity K_2 . When this is true, the 3 populations have almost the exact same general dynamics between the two models, as they reach a similar equilibrium state around month 100. This is shown in Figure 3.

There are some slightly more noticeable differences when the initial shark population is far from the carrying capacity of K_2 , however, this is unrealistic. The shark population in the Northern California coast is not increasing at a notably high rate, and the population is likely close to its carrying capacity (Kanive Et Al. 2021).

Once a sufficient amount of time had passed for the populations to become more stable, with less variation, around the 100 month mark, there were slight differences between the models. The otter population in Model 2 was slightly higher than Model 1, however, this was not a notable difference, and for our analysis is not of concern. Alongside this, the general dynamics of the populations are still the same.

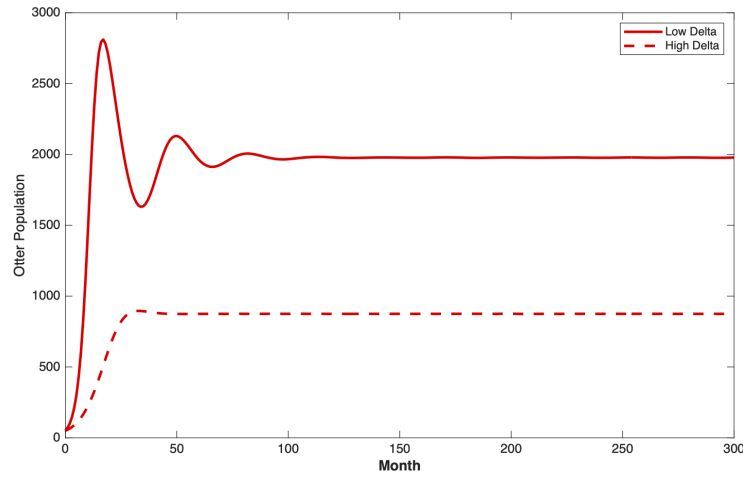
Model 2 may be more accurate for modeling shark population, as shark population has been increasing slowly in recent years; however, this increased accuracy is not beneficial, as the increase is too small to have any significant impact. Since Model 1 has 2 less parameters, and since Model 1 has only 2 state variables, Model 1 is much easier to analyze. Therefore, Model 1 will be much more beneficial for analysis, and will be the primary focus of the paper. Note that Model 2 followed very similar dynamics as Model 1 throughout testing, so it can be assumed what is said of Model 1 is true of Model 2.

b. Shark Predation Rate on Otters

The parameter of interest is the Shark-Otter Predation Coefficient (δ). This coefficient will give insight into the effect increased shark predation has on the model, including otter population dynamics and their survival. As δ increases, this increases the shark predation rate on otters δQ (In Model 1 Q is held constant at a value of 45), which represents increased predation on otters by sharks. This will help analyze the effect sharks are having on the survival of otter populations by examining dynamics of the otter population when δ is increased. The initial otter population ($N(0)$) is also of interest. Testing different initial otter populations will give insight into how large the otter population needs to be for a steady surviving population within model 1.

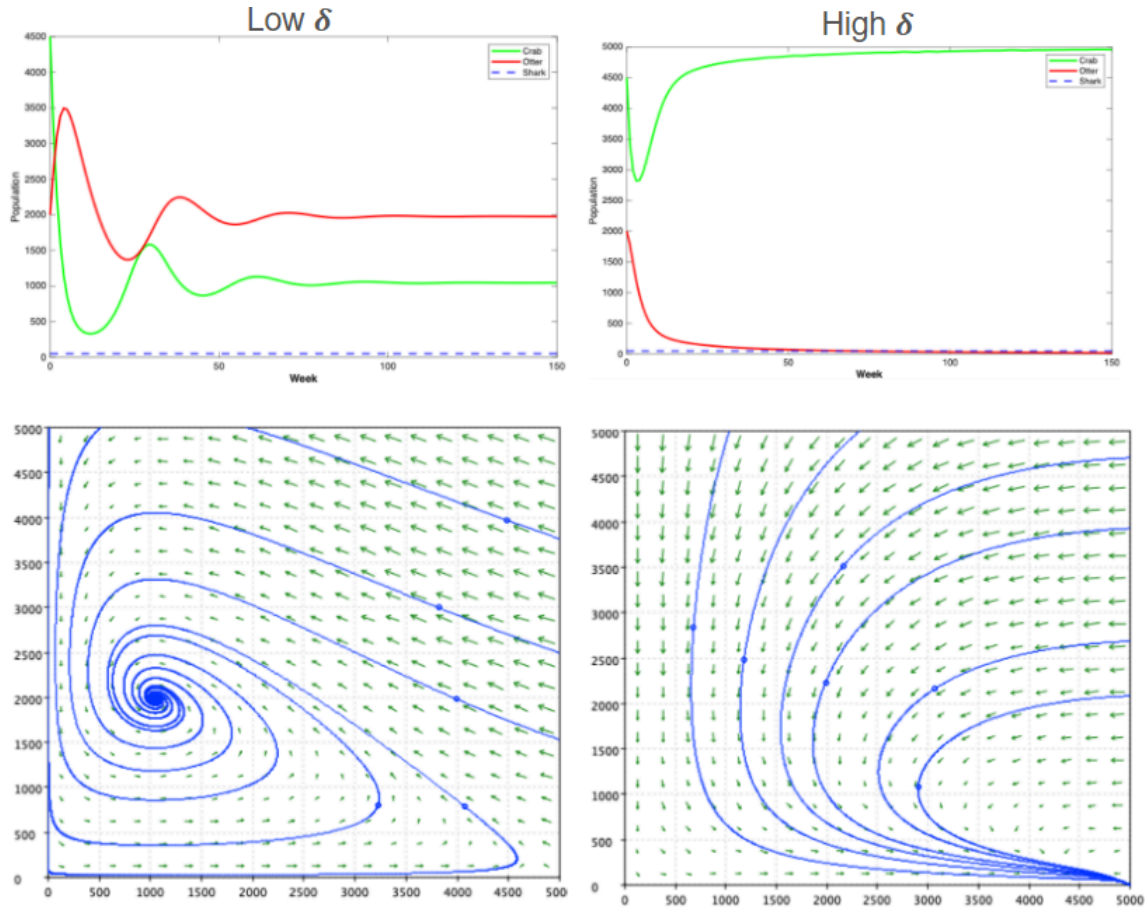
In the simulations, testing different initial otter populations has very little effect on their overall population dynamics. The effects of varying initial otter populations were apparent only in the transient behavior of the graph. With a larger value of $N(0)$, it takes longer for the initial large oscillations in their population to diminish, however, given a sufficient amount of time, the solution becomes the same as that of a smaller initial population. This was true for all nonzero initial otter populations tested. The lack of impact $N(0)$ has on the behavior of our model is a

Figure 4



Otter population graphed for a small value of $\delta=0.0001$ (solid) and a higher value of $\delta=0.005$ (dashed), with initial conditions $C(0)=4500$ and $N(0)=50$.

Figure 5



Model 1 simulated and otter and crab populations graphed temporally for a low value of $\delta=0.0001$ and very high value of $\delta=0.005$, with initial conditions $C(0)=4500$ and $N(0)=2000$. Below the temporal graphs is the corresponding phase plane

potential flaw of the model, as the initial nonzero value for N representing an otter population of 1 will still grow into a stable population within the model. This is unrealistic however, as a single otter cannot give rise to a stable population.

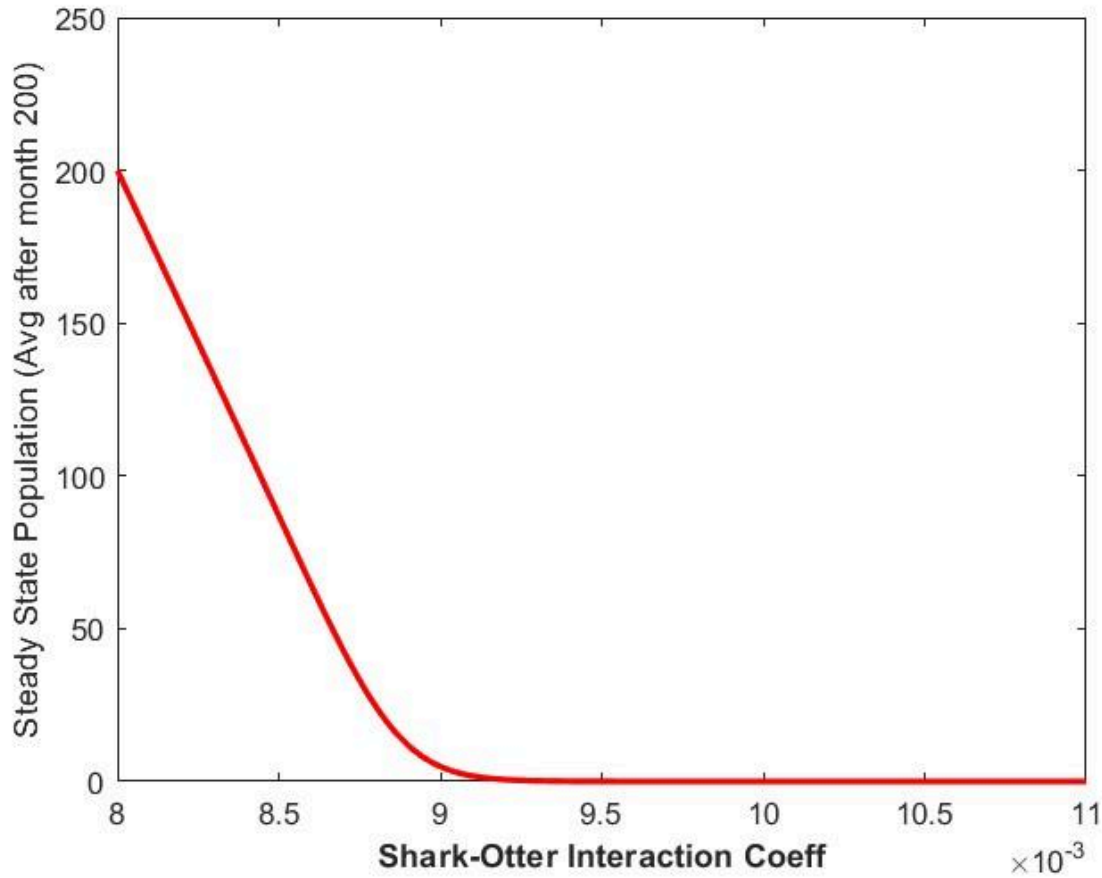
When δ is sufficiently low, for instance $\delta=0.0001$, the otter population oscillates around a nonzero value with decreasing amplitudes, as seen in the low delta curve in Figure 4. Further the otter population approaches the nonzero value after a sufficient amount of time has passed, around month 100. This is represented as a spiral approaching a fixed point when graphed in a phase plane, which is shown in Figure 5. This behavior in the model represents a surviving non-zero otter population and successful reintroduction of an otter population into an ecosystem with an existing shark population. When δ is increased by a small amount, $\delta=0.005$, the otter population exhibits the same behavior, but approaches a smaller value as seen in the high delta curve in Figure 4. This represents the otter population becoming steady at a smaller non zero number when shark predation is higher.

More notably, once δ becomes sufficiently large, the otter population stops oscillating around a steady nonzero value (population) with decreasing amplitudes, and monotonically drops to 0. In the phase plane for a large value of delta shown in Figure 5, the spiral is no longer seen, as the curves decrease into a node at approximately 0. This behavior represents the death of the otter population and failed reintroduction. Further, this fundamental change in behavior when varying δ is a bifurcation.

To further investigate this bifurcation, the values of δ for which this change in behavior occurs are of interest. To find them, a graph of the average otter population after the transient behavior in our simulation (about 200 months) is graphed against the value of δ for that run of the simulation. δ was varied from 0.008 to 0.011 with a step size of 0.00001. The resulting graph is shown in Figure 6. As seen in the graph, the otter population approaches 0 for $\delta > 0.0094$. So, for $\delta > 0.0094$, the reintroduced otter population does not survive. Further, the bifurcation occurs at δ when $0.009 < \delta < 0.0095$ in Model 1 with the chosen set of parameters. Note, shark population Q is held constant at $Q = 45$ for these runs. Therefore, for values of the predation rate of sharks on otters $\delta Q > 0.423$ the population approaches 0, and for $0.405 < \delta Q < 0.428$ the bifurcation occurs.

This bifurcation implies that the shark-otter predation coefficient δ has a significant impact on the long term survival of otters within the model. Since δQ represents the rate of shark predation on otters, this tells that our model suggests the rate of shark predation can fundamentally change population dynamics of the otter population being reintroduced in an area. Within the model, when shark predation is sufficiently low, the otter population approaches a non-zero value, representing the survival of a reintroduced otter population. This can be seen in the spiral in Figure 5 under low δ . When shark predation becomes too high, the otter population will drop to zero, implying the otter population doesn't survive. This can be seen in Figure 5 under high δ . This supports the hypothesis that shark predation on otters in Northern California (where there are higher rates of shark predation) has a significant impact on the survival of otter populations in that area.

Figure 6



*Average otter population after 200 months in a run of model plotted against the value of δ in that run.
 $N(0) = 50$ and $C(0) = 4500$*

It should be noted that the crab population never went to 0 in any runs of the simulation. In order to do this, unrealistic parameters were needed. Within the model, this implies that crab populations would not be in danger if otters were to be reintroduced.

4. Conclusions

The results of this study point to several main conclusions. First, there was no meaningful difference between modeling the shark population with logistic growth and assuming it remained constant. Since shark numbers in Northern California are likely already close to their carrying capacity, this similarity in outcomes suggests that the simpler model is sufficient for examining long-term system behavior.

Second, the model clearly shows that changes in shark predation can fundamentally alter otter population dynamics. When predation levels stayed low, otter numbers eventually settled into a stable, nonzero equilibrium after some initial oscillations. However, once predation was increased past a narrow threshold the otter population collapsed to zero, which would cause a

reintroduction effort to fail. These findings support the idea that shark predation does indeed play a major role in determining whether reintroduced otter populations can survive in a new ecosystem.

Third, varying the initial otter populations did not influence the long-term outcomes of the model, as long as the initial population stayed above zero. This is a major limitation of the current model because it essentially allows a single otter to grow into a stable population, which is not realistic. Adding an extinction threshold, such as an Allee effect or a requirement for the presence of both female and male otters, would make the model more biologically accurate. For this reason, no conclusions can be made about the minimum number of otters needed for a successful reintroduction.

In addition, incorporating seasonal variation would likely improve the model. Otters have periods of higher birth rates during the year, and shark presence and predation intensity peak from late summer to early fall in Northern California. Including these seasonal patterns could affect the model's transient dynamics and could potentially show that the timing of the reintroduction influences its success.

Lastly, the model did not show a collapse of the crab prey population unless unrealistic parameter values were used. This suggests that crab populations are unlikely to be overexploited under normal conditions, but because this part of the model was not analyzed in depth, further work is needed to understand when major crab declines might occur.

While the simplified model highlights the strong influence of shark predation on otter reintroduction outcomes, several improvements such as adding demographic thresholds, seasonal processes, and a more detailed analysis of prey dynamics would make future versions more realistic and more useful for conservation planning.

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